Problem Set N – Template

4SC000 Optimal Control and Reinforcement Learning

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Problem T.1 Consider a 2-stage inventory problem with state $x_k \in \{0, 1, 2\}$ (units in stock) and decision $u_k \in \{0, 1\}$ (order 0 or 1 unit), horizon $N = 2$. The dynamics are $x_{k+1} = \min(x_k + u_k, 2)$, and the stage cost is $g_k(x_k, u_k) = u_k + (2 - x_k)$ (ordering cost plus a shortage penalty), with terminal cost $J_2(x_2) = 0$. Given $x_0 = 0$, compute the optimal policy $\mu_0^\star, \mu_1^\star$ and the optimal cost using the DP algorithm.

Solution

Backward recursion, $J_2(x_2) = 0$ for all x_2.
Stage 1 ($x_1 \in \{0, 1, 2\}$):

$$J_1(x_1) = \min_{u_1 \in \{0,1\}} \; u_1 + (2 - x_1) + J_2(\min(x_1 + u_1, 2))$$

Since $J_2 \equiv 0$, the minimizing choice is always $u_1 = 0$ (ordering never helps with one stage left and no terminal cost), giving $J_1(x_1) = 2 - x_1$, $\mu_1^\star(x_1) = 0$.
Stage 0 ($x_0 = 0$):

$$J_0(0) = \min_{u_0 \in \{0,1\}} \; u_0 + 2 + J_1(\min(u_0, 2))$$

- $u_0 = 0$: cost $= 0 + 2 + J_1(0) = 0 + 2 + 2 = 4$ - $u_0 = 1$: cost $= 1 + 2 + J_1(1) = 1 + 2 + 1 = 4$
Both are optimal here (tie); take $\mu_0^\star(0) = 1$ to end stage 0 with more stock. Optimal cost $J_0(0) = 4$.